

VCTK 1000-Pair Analysis

In-Distribution Evaluation Report

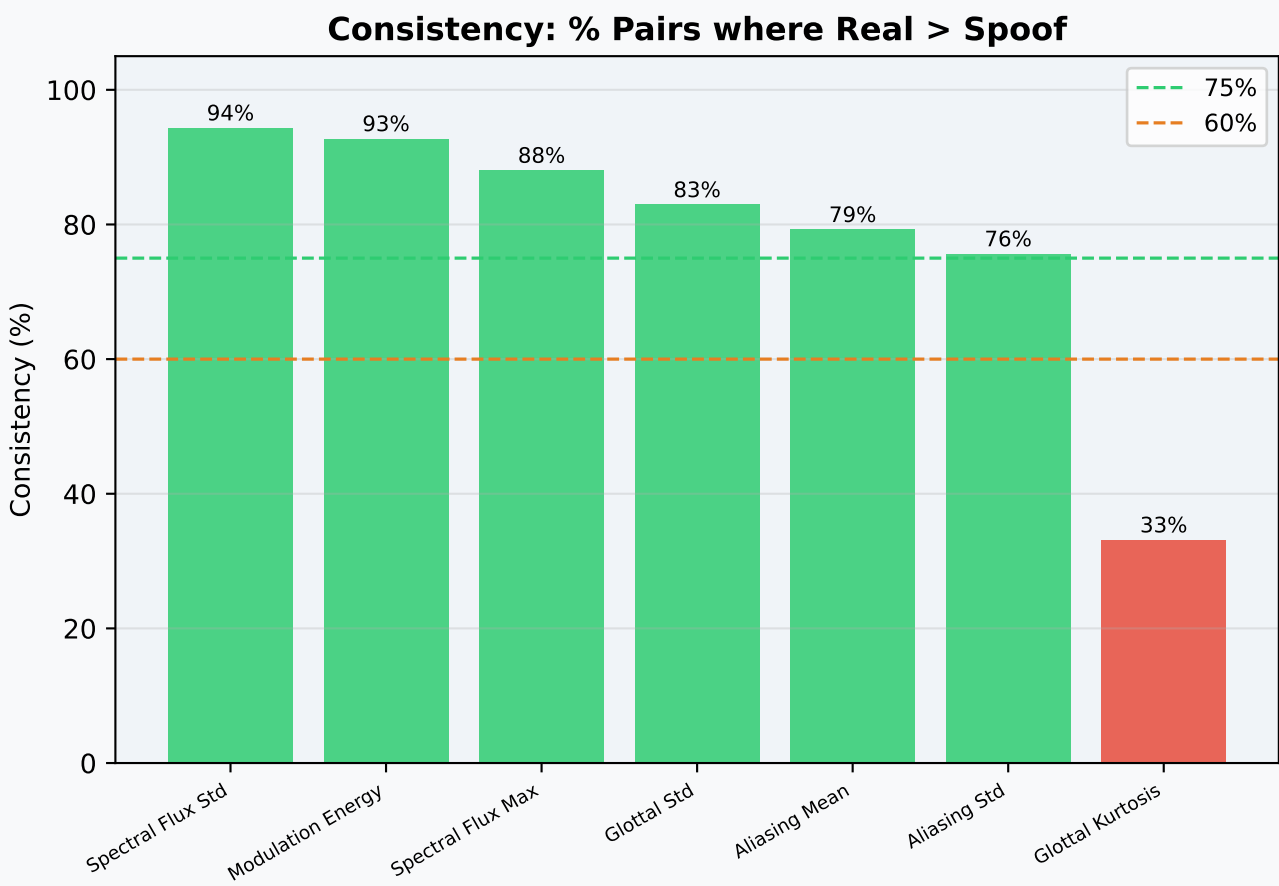
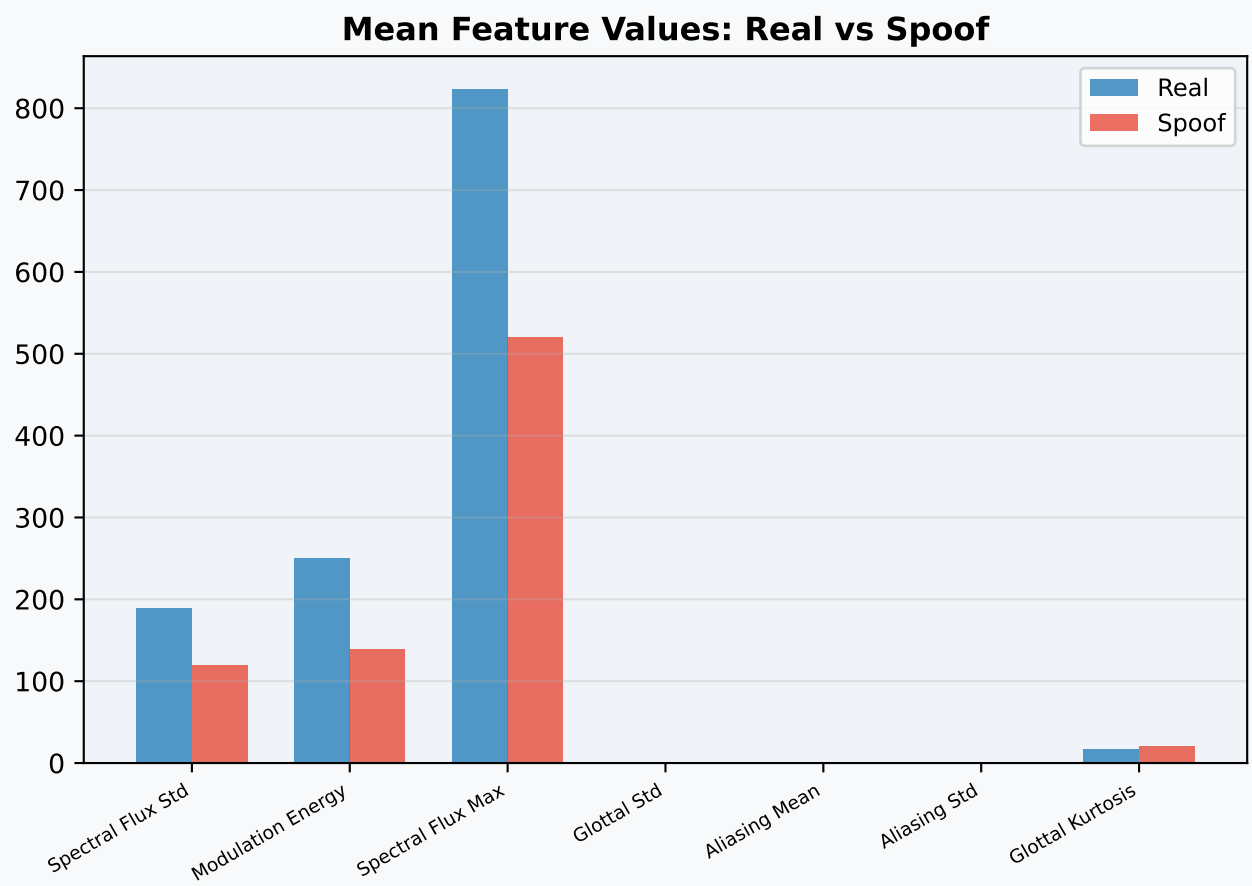
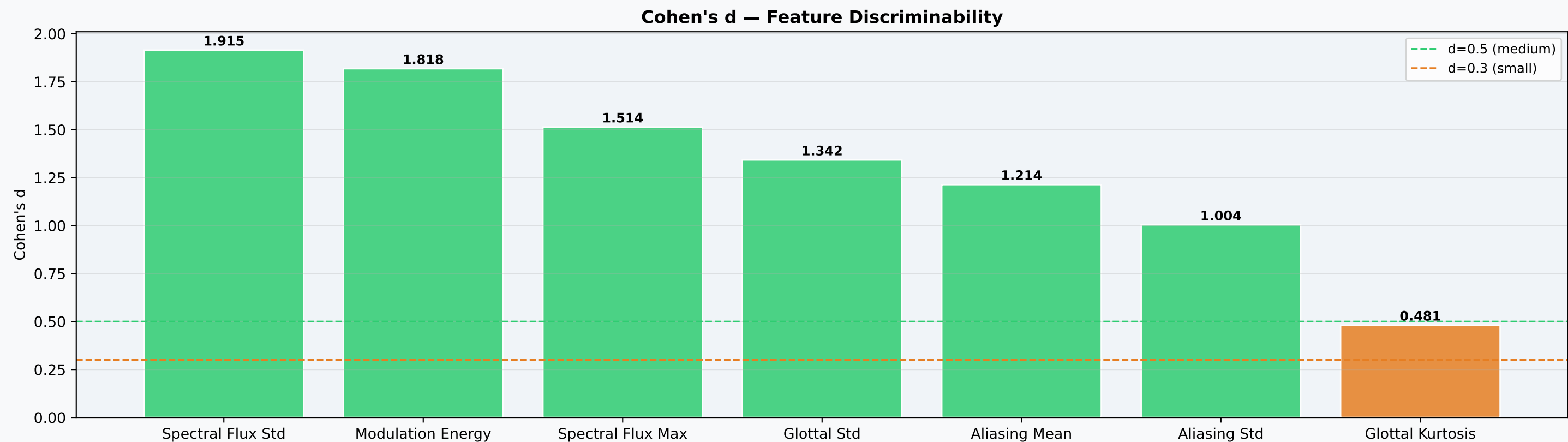
EE 123 Project — Ved, Ching, Erick

Real: 1000 files (source-vctk1k)

Spoof: 1000 files (converted-vctk1k)

Models trained on source vs converted (1000 vs 1000, balanced)

Feature Discriminability — VCTK 1000 Pairs



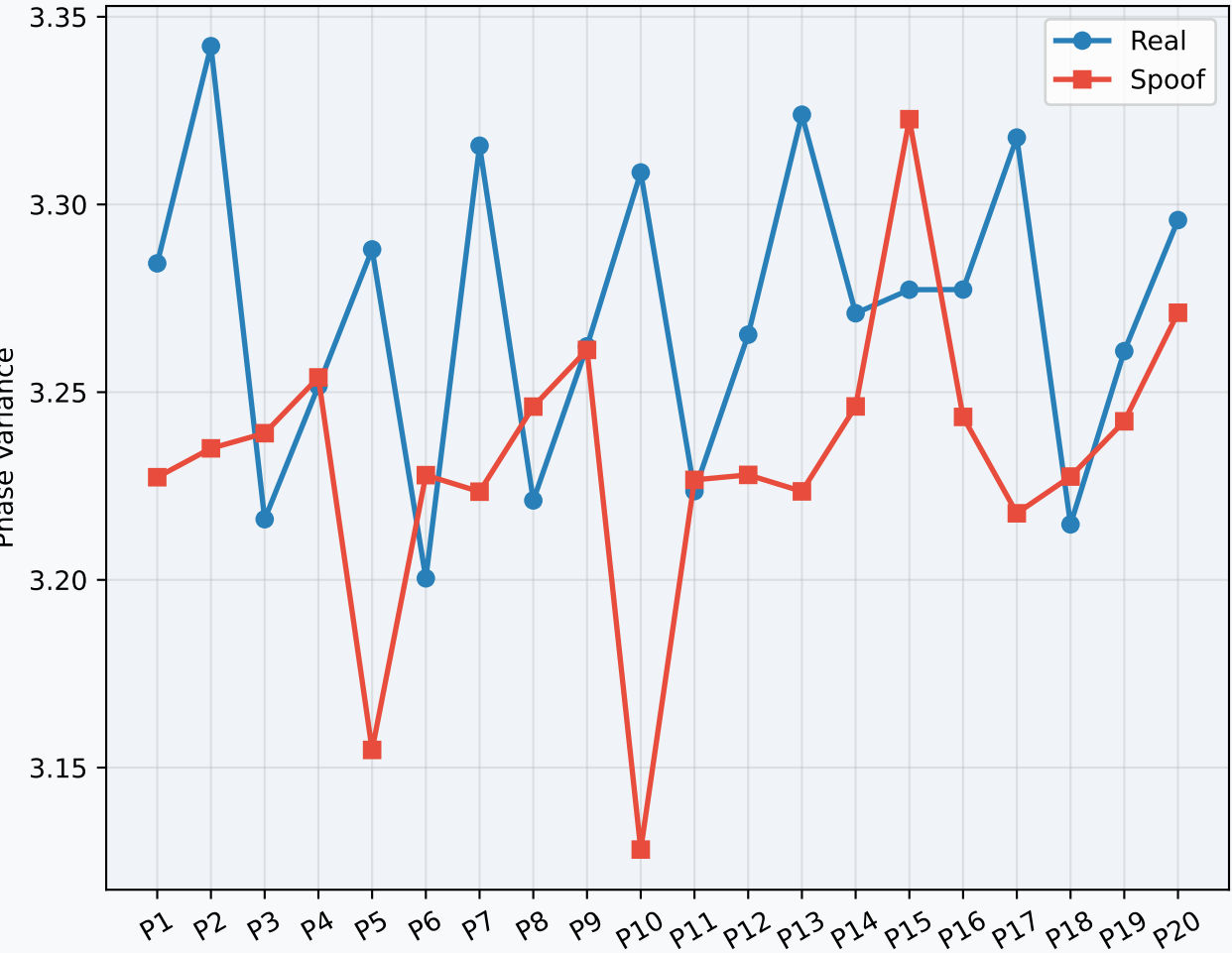
VCTK 1000-Pair Aggregate Feature Statistics

Feature	Real Mean	Real Std	Spoof Mean	Spoof Std	Diff%	Cohen's d	Real>Spoof (out of 1000 pairs)	Consist%	Keep?
Spectral Flux Std	189.228	47.841	119.153	19.744	+37.0%	1.915	943/1000	94.3%	✓ KEEP
Modulation Energy	249.936	80.930	138.952	30.066	+44.4%	1.818	927/1000	92.7%	✓ KEEP
Spectral Flux Max	822.462	254.223	520.393	122.480	+36.7%	1.514	880/1000	88.0%	✓ KEEP
Glottal Std	0.023	0.008	0.015	0.004	+35.5%	1.342	829/1000	82.9%	✓ KEEP
Aliasing Mean	0.208	0.108	0.106	0.049	+49.0%	1.214	792/1000	79.2%	✓ KEEP
Aliasing Std	0.299	0.160	0.173	0.077	+42.2%	1.004	756/1000	75.6%	✓ KEEP
Glottal Kurtosis	16.562	7.461	20.699	9.612	-25.0%	0.481	331/1000	33.1%	✗ DROP

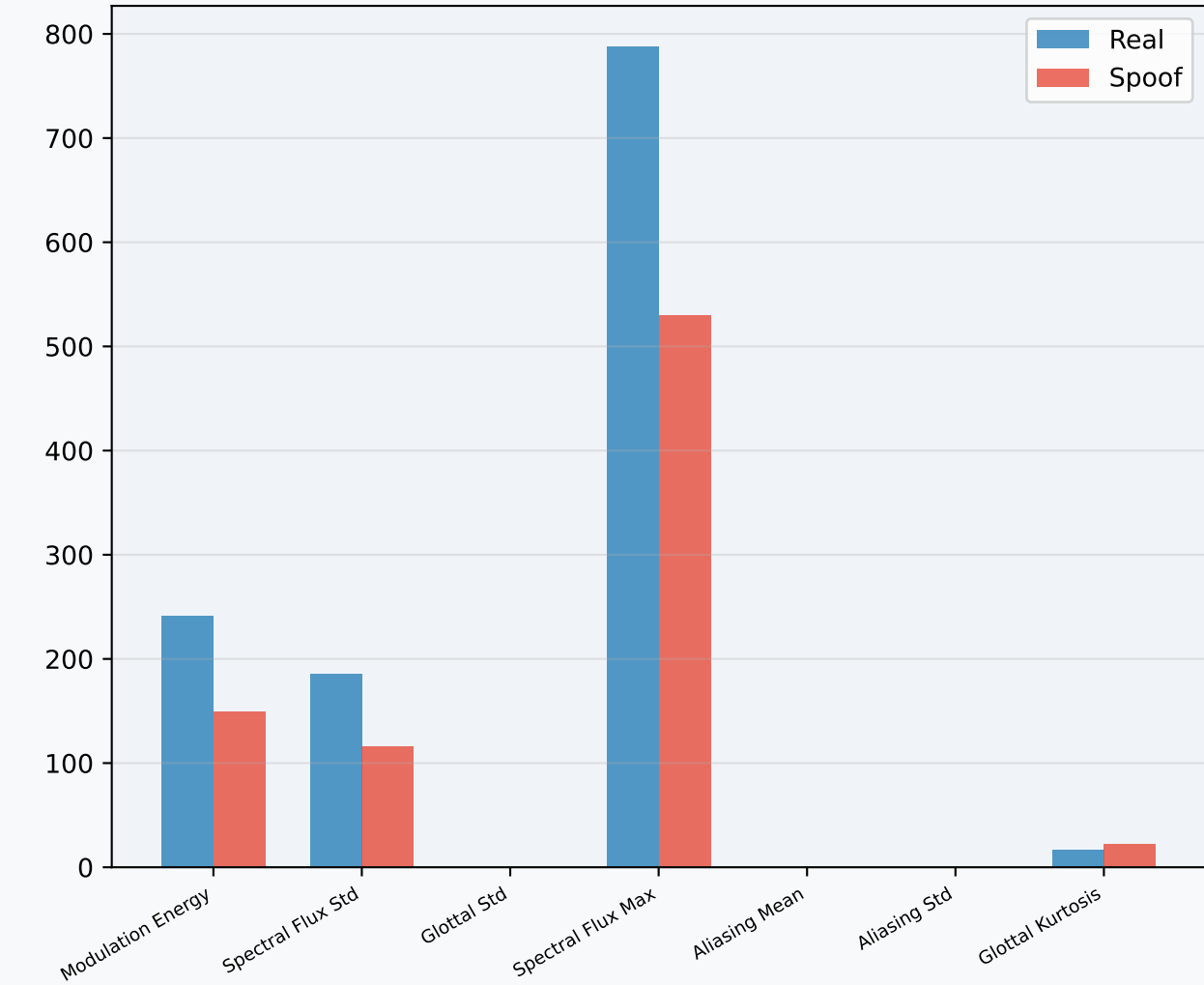
Green = strong ($d>0.5$) | Yellow = moderate ($d>0.3$) | Red = weak/drop

Signal Analysis Summary — 20 Sampled Pairs

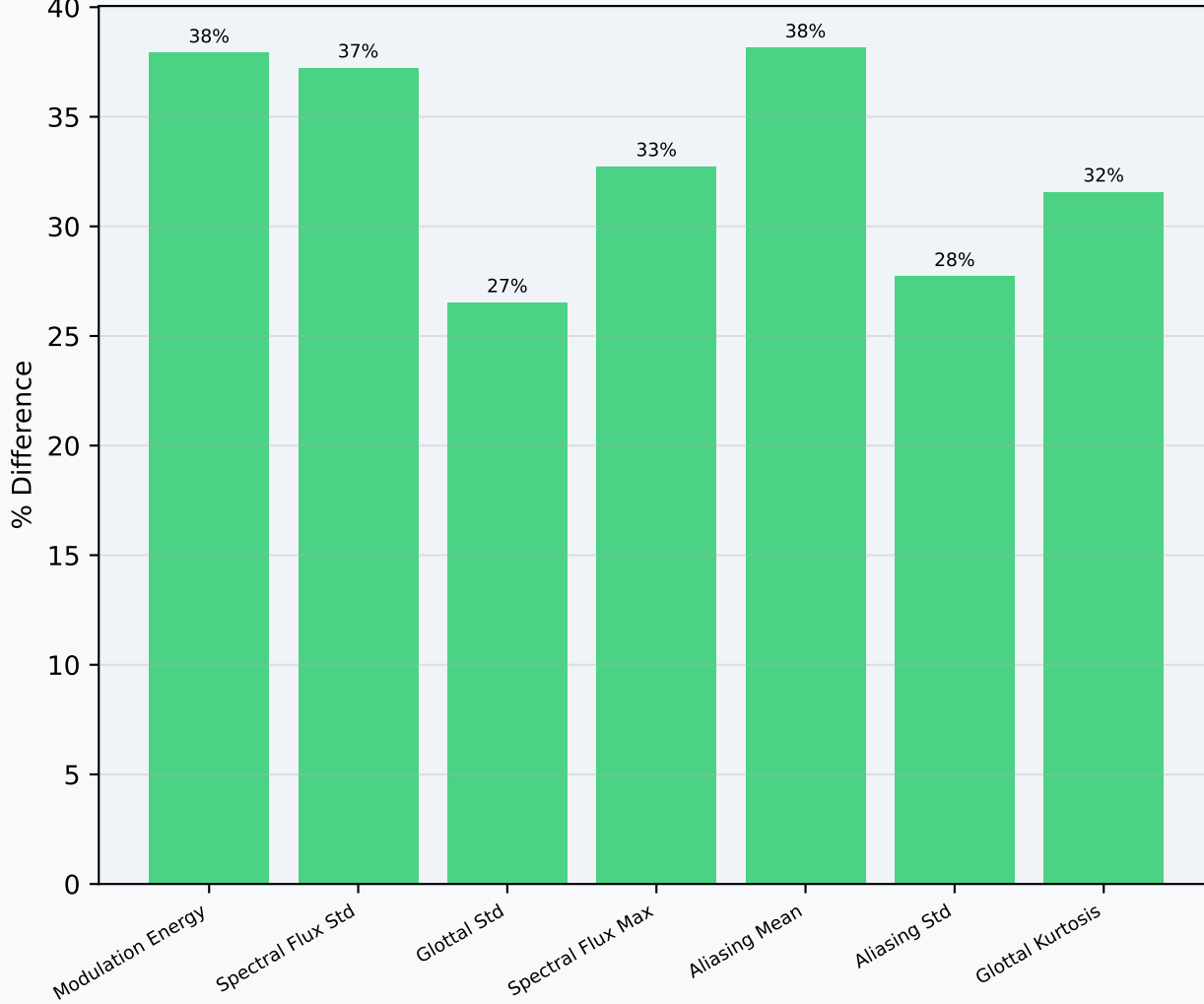
Mean Phase Variance per Pair



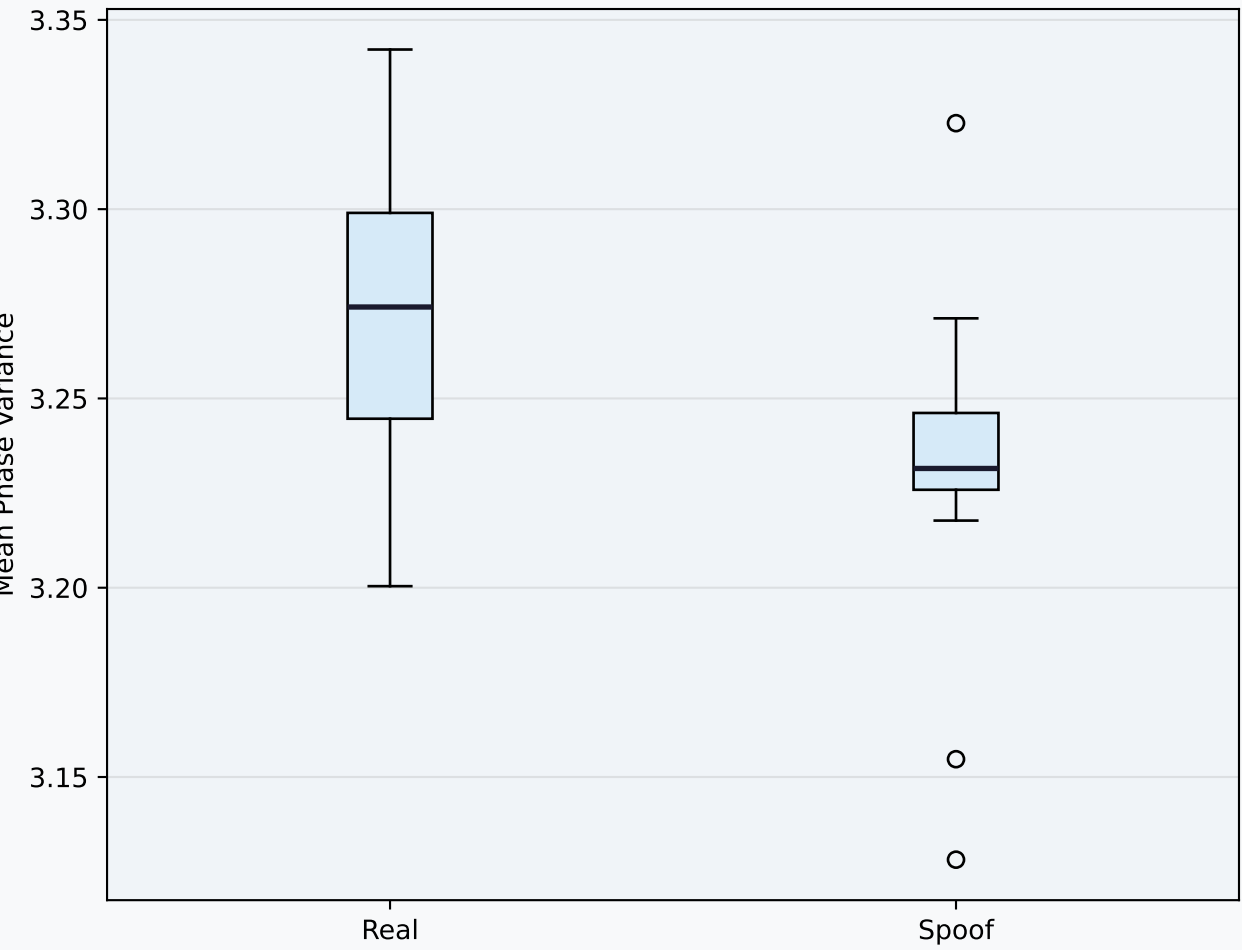
Avg Physics Features



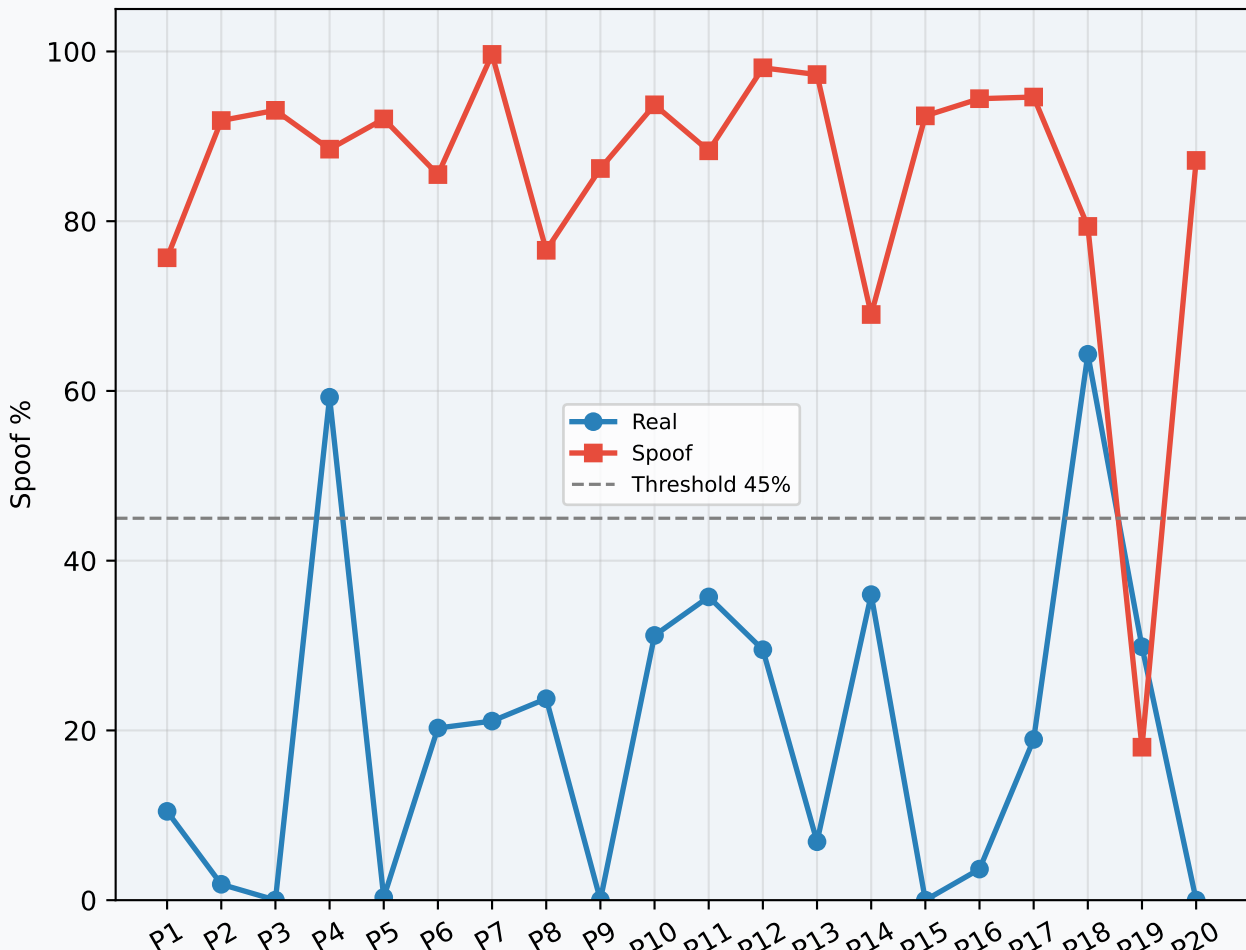
Avg % Difference per Feature



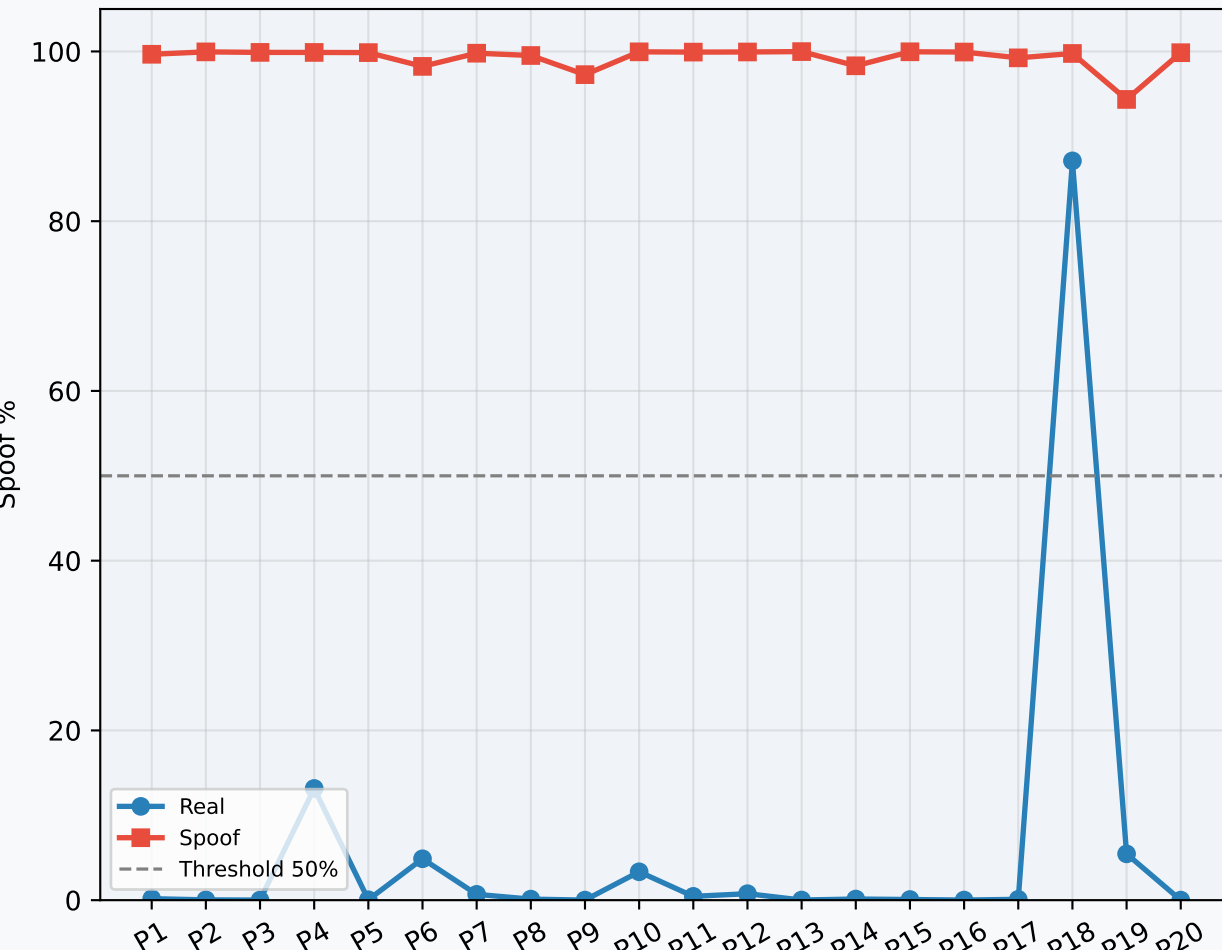
Phase Variance Distribution



LR Spoof Probability per Pair

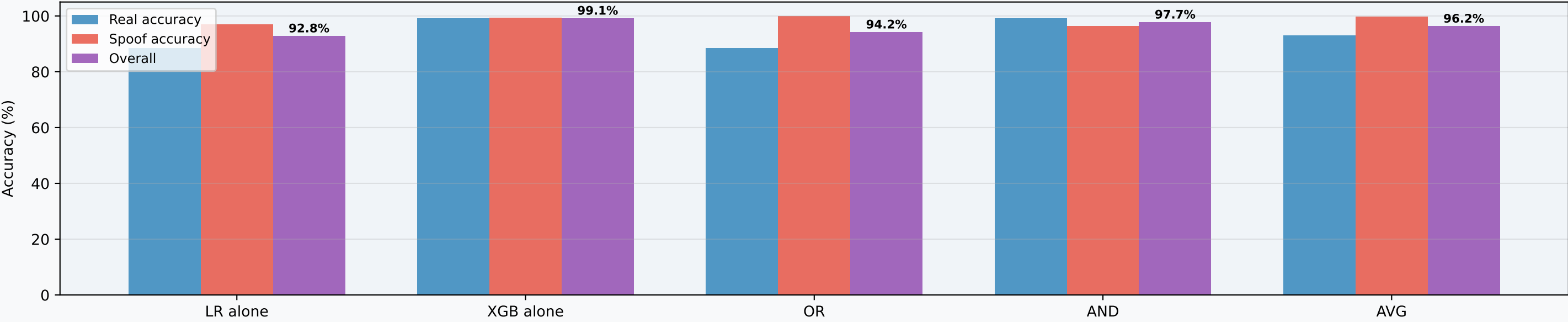


XGB Spoof Probability per Pair

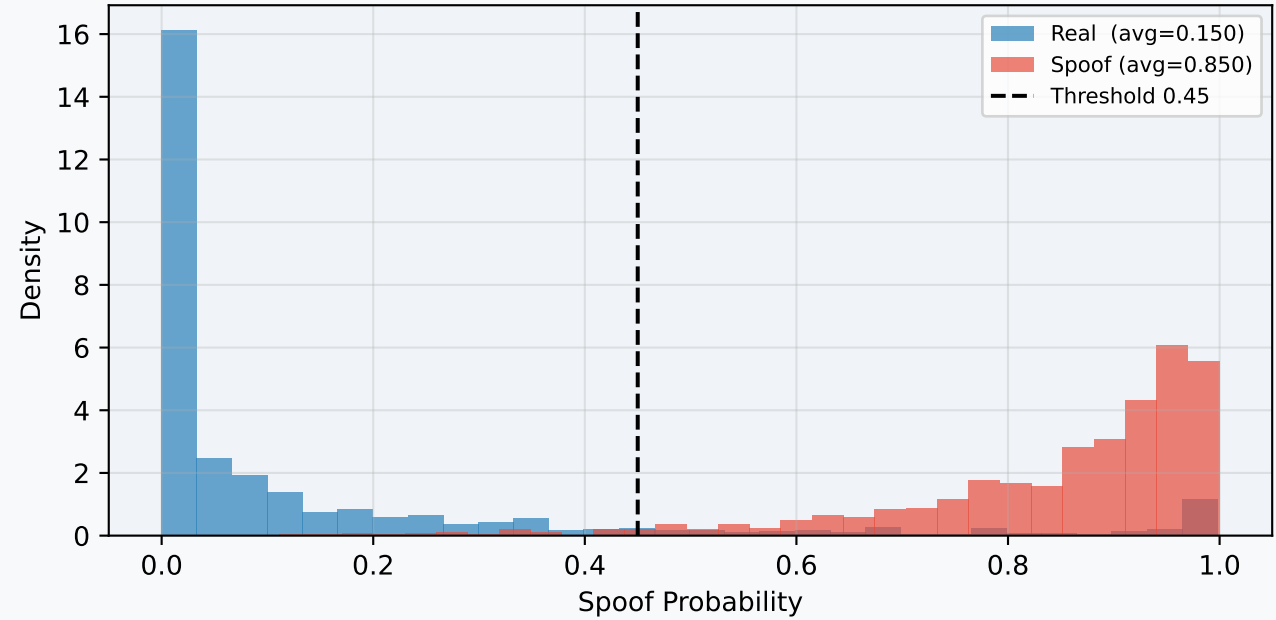


VCTK 1000-Pair — Ensemble Evaluation Results

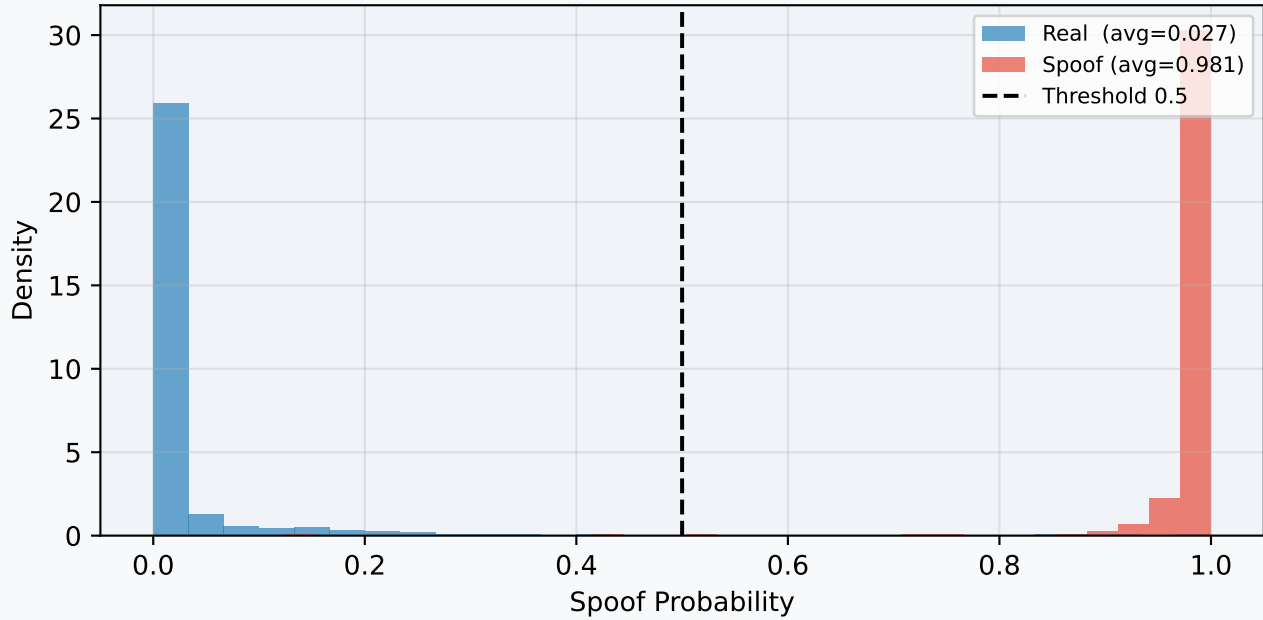
Accuracy by Strategy



LR Probability Distribution



XGBoost Probability Distribution



Raw Count Summary

Strategy	Real Correct	Real Total	Real%	Spoof Correct	Spoof Total	Spoof%	Overall Correct	Overall Total	Overall%
LR alone	885	1000	88.5%	970	1000	97.0%	1855	2000	92.8%
XGB alone	990	1000	99.0%	992	1000	99.2%	1982	2000	99.1%
OR	885	1000	88.5%	998	1000	99.8%	1883	2000	94.2%
AND	990	1000	99.0%	964	1000	96.4%	1954	2000	97.7%
AVG	929	1000	92.9%	996	1000	99.6%	1925	2000	96.2%